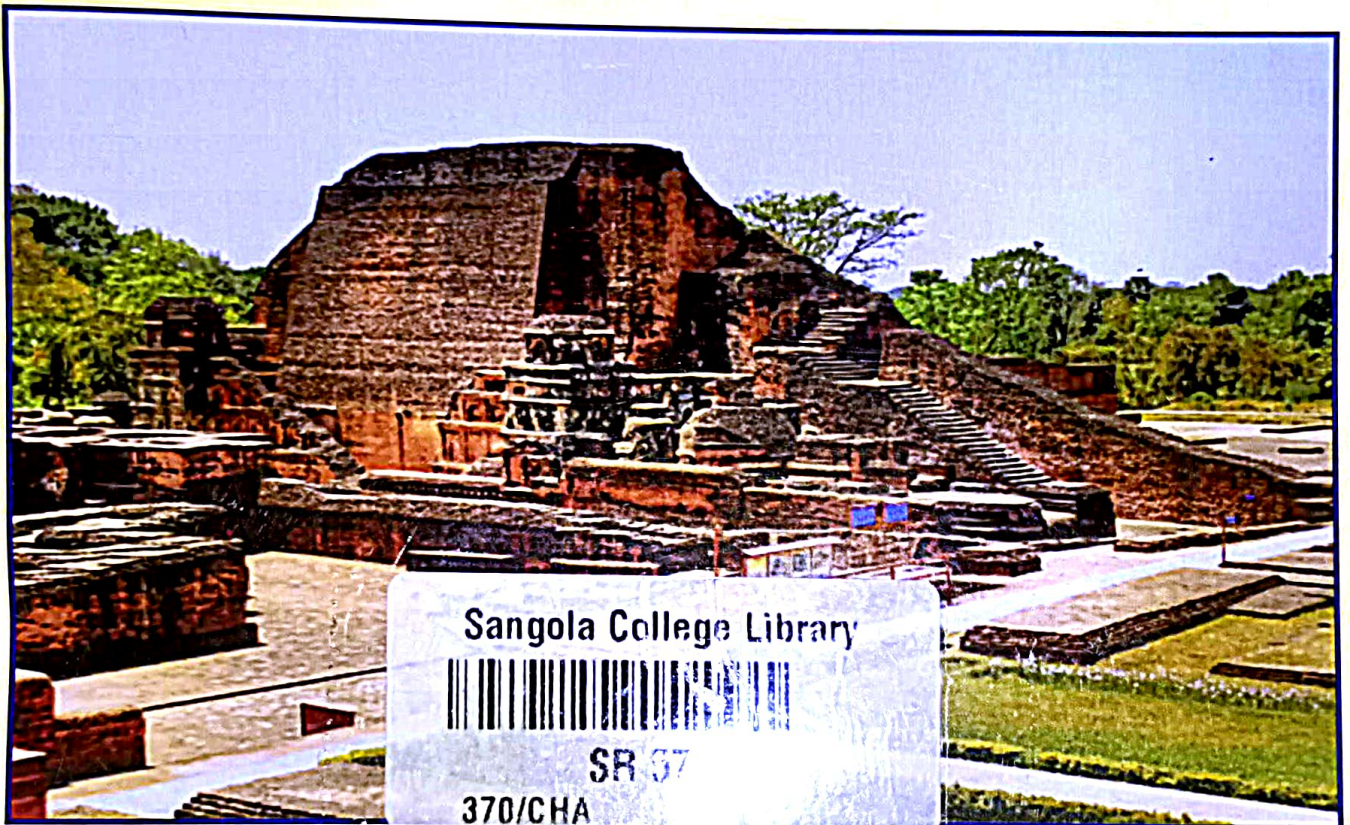


As per New NEP-2020 Pattern Syllabus of
Punyashlok Ahilyadevi Holkar,
Solapur University

B.Sc. & B.Sc. (ECS) Part-I : Semester-I

INTRODUCTION TO INDIAN KNOWLEDGE SYSTEM



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SYLLABUS

Unit I: Bhāratīya Civilization and Development of Knowledge System (10 Hours)

- 1.1 **Indian Knowledge Systems (IKS):** Foundational concepts and characteristics of Indian philosophical and scientific traditions.
- 1.2 **Significance of Ancient Knowledge:** Understand the need of ancient knowledge and its role in shaping modern civilization.
- 1.3 **Indian Philosophical Systems:** Development and distinctive features of both vedic and non-vedic philosophical systems, analyzing their historical and cultural contexts.
- 1.4 **Ancient Educational Institutions:** Takṣaśilā and Nālandā, and notable alumni who contributed to the spread of knowledge.

Unit II: Science and Mathematics (10 Hours)

- 2.1 **Physics:** Indian concepts related to the atom, laws of motion, electricity, magnetism, and the mystery of light.
- 2.2 **Chemistry:** Indian innovations in daily-life chemistry, including dyes, paints, cements, glass, pottery, and metallurgy.
- 2.3 **Mathematics:** Indian origin of foundational, mathematical and statistical concepts, such as the concept of zero, pi, the decimal number system and Probability, Bodhayana Sutra and Vedic Mathematics, emphasizing their influence on mathematical thinking.

Unit III: Life, Environment, and Health (10 Hours)

- 3.1 **Approaches to Life and Health:** Traditional Indian perspectives on life sciences, including plant science, anatomy and physiology, and agriculture.
- 3.2 **Environmental Awareness:** India's traditional emphasis on ecology and environmental sustainability. Ancient practices that promote environmental stewardship.

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*** Practice Question Paper**

Q.1 - Q.2

Punyashlok Ahilyadevi Holkar Solapur University, Solapur
Nature of Question Paper for NEP-2020 Pattern
(w.e.f. June 2024)

Time :

Total Marks 30

Instructions :

1. All questions are compulsory.
2. Figures to the right indicate full marks.

Q. 1 Multiple Choice Questions :

(06)

1.
(a) (b) (c) (d)
2. ...do...
3. ...do...
4. ...do...
5. ...do...
6. ...do...

Q. 2 Answer the following (Any three) :

(06)

- (i)
- (ii)
- (iii)
- (iv)
- (v)

Q. 3 Answer the following (Any two) :

(06)

- (i)
- (ii)
- (iii)

Q. 4 Answer the following (Any two) :

(06)

- (i)
- (ii)
- (iii)

Q. 5 Answer the following (Any one) :

(06)

- (i)
- (ii)

BHARATIYA CIVILIZATION AND DEVELOPMENT OF KNOWLEDGE SYSTEM

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1.1 INTRODUCTION

Bhāratīya (Indian) civilization, one of the oldest and most enduring in the world, stands as a testament to human ingenuity, resilience, and cultural richness. Spanning over 5,000 years, it has been a cradle of knowledge, spirituality, and artistic expression, influencing the course of human history in profound ways. At the core of this civilization lies a deep philosophical tradition, rooted in ancient texts such as the Vedas and Upanishads. These works explore the nature of existence, the self, and the universe, offering timeless concepts like *Brahman* (universal consciousness), *Atman* (individual soul), and *Moksha* (liberation). Spiritual practices like yoga and meditation, originating from these teachings, have gained global recognition for their ability to harmonize body, mind, and soul.

Indian civilization has also been a trailblazer in science, mathematics, and medicine. The astronomer Brahmagupta introduced the concept of zero, revolutionizing mathematics and computation worldwide. "Aryabhata's" contributions to astronomy, such as the Earth's rotation and the calculation of planetary orbits, were centuries ahead of their time. In medicine, Sushruta, known as the "Father of Surgery," authored the *Sushruta Samhita*, detailing surgical techniques and principles of Ayurveda, India's holistic health system. The ancient grammarian Panini, through his masterpiece *Ashtadhyayi*, set a standard for linguistic precision in Sanskrit, a language that became a medium for transmitting profound knowledge and influenced many languages across Asia.

The literary and artistic heritage of Bhāratīya civilization is equally remarkable. Epics such as the Mahabharata and Ramayana are not just stories but repositories of ethical principles, governance, and human values. The Bhagavad Gita, a part of the Mahabharata, continues to inspire readers with its teachings on duty, detachment, and the pursuit of self-realization. Indian art and architecture, from the intricate carvings of Khajuraho to the majestic stupas of Sanchi, reflect a harmonious blend of beauty and symbolism. Iconic structures like the Taj Mahal exemplify

India's architectural genius and cultural synthesis. Similarly, classical dance forms like Bharatanatyam and Kathak, along with Hindustani and Carnatic music, convey spiritual narratives and cosmic rhythms, enriching the cultural tapestry of the world.

One of the most enduring aspects of Indian civilization is its emphasis on universal values and interconnectedness. Principles like *Ahimsa* (non-violence), *Dharma* (righteousness), and *Vasudhaiva Kutumbakam* ("the world is one family") reflect a vision of compassion, inclusion, and harmony. These values have inspired global movements for social justice and peace, most notably through the teachings of Mahatma Gandhi. The civilization also emphasizes the interdependence of humans and nature, encouraging sustainable living and ecological balance.

The contributions of Bhāratīya civilization extend to the modern world, with its ancient wisdom continuing to shape contemporary practices. Yoga and meditation, now globally recognized, contribute to mental and physical well-being, while its emphasis on ethical living and balance offers guidance in a rapidly changing world. By exploring this civilization, we not only appreciate its unparalleled achievements but also draw inspiration from its timeless teachings, which remind us to harmonize material progress with spiritual growth and ethical integrity.

1.2 INDIAN KNOWLEDGE SYSTEMS (IKS): FOUNDATIONAL CONCEPTS AND CHARACTERISTICS

Indian Knowledge Systems (IKS) are a collection of ideas and knowledge that developed in India over thousands of years. These systems cover many areas, such as philosophy, science, medicine, art, and spirituality. They are deeply connected to Indian culture and traditions. Below are the main ideas and features of Indian philosophical and scientific traditions explained in simple terms.

1.2.1 Main Ideas of Indian Knowledge Systems

- (a) **Connection between knowledge and spirituality:** In Indian traditions, knowledge is not just about learning facts. It is also

about understanding life, the universe, and spiritual truths. The ultimate goal is to achieve self-realization and liberation (*moksha*).

- (b) **Everything is connected:** Indian knowledge sees the world as an interconnected whole. Science, philosophy, art, and ethics are not treated as separate subjects but are combined to create a complete understanding of life.
- (c) **Searching for truth:** Indian traditions focus on finding the truth (*satya*). This search is done through observation, reasoning, and self-awareness. Knowledge is seen as a journey, not just a destination.
- (d) **Oral and written traditions:** In the past, Indian knowledge was passed down orally from teachers to students through the *guru-shishya* system. Later, it was written down in texts like the Vedas, Upanishads, and epics such as the Mahabharata and Ramayana.
- (e) **Time is cyclical:** Indian philosophy believes that time moves in cycles, unlike the Western view of time as a straight line. Concepts like *yugas* (ages) and *kalpas* (cosmic cycles) explain this cyclical nature of time.

1 *yuga* = 4.32 million years

1 *kalpa* = 4.32 billion years

- (f) **Knowledge and ethics go together:** In Indian traditions, knowledge is always guided by ethics (*dharma*). This means that knowledge should be used responsibly to help people and maintain harmony in society and nature.

1.2.2 Features of Indian Philosophical Traditions

- (a) **Different schools of thought:** Indian philosophy has many schools of thought, such as Nyaya, Yoga, Vedanta, and Buddhism. Each has its own ideas, but all aim to understand life, the mind, and the universe.
- (b) **Focus on self-discovery:** Indian philosophy emphasizes self-awareness and personal experience. Practices like meditation and

yoga help individuals understand themselves and their connection to the world.

- (c) **Logic and reasoning:** Reasoning and debate are important in Indian philosophy. For example, the Nyaya school focuses on logic and clear thinking to understand the truth.
- (d) **Open to different ideas:** Indian traditions welcome a variety of views and adapt to new situations. This flexibility has kept Indian knowledge alive and relevant for thousands of years.

1.2.3 Features of Indian Scientific Traditions

- (a) **Observation and experimentation:** Indian sciences, such as Ayurveda (medicine) and astronomy, are based on careful observation and practical experiments. For example, Aryabhata made accurate calculations about the Earth and planets, and Sushruta was an expert in surgery.
- (b) **Mixing science with philosophy:** Indian scientific traditions often include philosophical ideas. For instance, Ayurveda uses the concept of the five elements (*panchamahabhutas*) and the three doshas to understand health and diseases.
- (c) **Harmony with nature:** Indian sciences focus on living in balance with nature. Ancient farming methods, water conservation, and architecture (like Vastu Shastra) are examples of this eco-friendly approach.
- (d) **Mathematics and innovation:** Indian mathematicians introduced important concepts like zero, the decimal system, and advanced geometry. Their work influenced mathematics worldwide.
- (e) **Focus on health and balance:** Indian medicine, especially Ayurveda, looks at health as a balance between body, mind, and soul. It also emphasizes prevention and natural remedies to maintain well-being.

1.2.4 Why Indian Knowledge Systems are Important Today?

Indian knowledge systems provide timeless wisdom that is still useful today. They teach us to live in harmony with nature, take care of our

health, and balance material and spiritual growth. They also show us how to combine different fields of knowledge to solve problems.

By studying these systems, we can learn from the past and find better ways to live in today's fast-changing world.

1.3 SIGNIFICANCE OF ANCIENT KNOWLEDGE

Understanding its importance and role in shaping modern civilization:

Ancient knowledge refers to the wisdom, discoveries, and ideas developed by early civilizations like India, Mesopotamia, Egypt, Greece, and China. This knowledge, built over centuries, laid the foundation for modern science, technology, philosophy, art, and culture. Below is a detailed explanation of why ancient knowledge is important and how it has shaped the modern world.

1.3.1 Importance of Ancient Knowledge

(a) Foundation of modern sciences:

Ancient civilizations made significant contributions to fields like mathematics, medicine, astronomy, and engineering. For example:

- Indian mathematicians introduced the concept of zero and the decimal system, which are essential in modern mathematics and technology.
- Ancient Greek thinkers like Pythagoras and Archimedes laid the groundwork for geometry and mechanics.
- Chinese inventors developed technologies like paper and the compass, which transformed communication and navigation.

(b) Understanding nature and sustainability:

Ancient knowledge systems focused on living in harmony with nature. Practices like organic farming, water conservation, and natural medicine reflect a deep understanding of ecological balance. These ideas are now being revisited to address modern challenges like climate change and environmental degradation.

(c) Philosophy and ethics:

Ancient knowledge systems provided valuable lessons in philosophy and ethics, guiding how humans should live and interact with one another. Indian texts like the Bhagavad Gita and Greek works by Plato and Aristotle explore morality, leadership, and the purpose of life, which are still studied today.

(d) Cultural and artistic heritage:

Ancient civilizations created art, literature, and architecture that continue to inspire modern creativity. For example, the epics of the Mahabharata and the Iliad are timeless stories of human emotion and values. The architectural wonders like the pyramids and temples demonstrate advanced engineering skills.

(e) Preservation of human history:

Ancient texts, inscriptions, and artifacts help us understand how early societies lived, thought, and evolved. This knowledge connects us to our roots and teaches us how civilizations grew and adapted over time.

1.3.2 Role of Ancient Knowledge in Shaping Modern Civilization**(a) Development of scientific thinking:**

Ancient knowledge fostered curiosity and inquiry, which are essential for scientific progress. Early scientists observed nature, experimented, and recorded their findings. Their work paved the way for modern research and technological innovation. For instance:

- Aryabhata's astronomical calculations influenced the study of space and time.
- Hippocrates' work in ancient Greece established the foundation of modern medicine.

(b) Contribution to governance and law:

Ancient civilizations developed systems of governance and law that influenced modern political systems. For example:

- The Indian concept of *dharma* emphasized justice and responsibility.
- The Roman legal system inspired many modern legal codes.

(c) Advancement in architecture and urban planning:

Ancient knowledge shaped the design of cities, buildings, and infrastructure.

- The Harappan civilization of India demonstrated advanced urban planning, with well-organized cities, drainage systems, and granaries.
- Roman aqueducts and roads set standards for public infrastructure.

(d) Inspiration for modern arts and literature:

Ancient myths, legends, and stories continue to inspire modern literature, films, and art. Epics, sculptures, and paintings from ancient cultures preserve universal themes of love, courage, and morality that resonate even today.

(e) Revival of ancient practices for modern problems:

Modern societies are re-discovering ancient practices to solve current challenges. For example:

- Yoga and Ayurveda are ancient Indian practices gaining global recognition for promoting health and well-being.
- Traditional agricultural methods are being studied for sustainable farming.

1.3.3 Lessons from Ancient Knowledge for Modern Times

(a) Balance between science and ethics: Ancient systems remind us that knowledge should be guided by ethics and used responsibly for the betterment of humanity.

(b) Respect for nature: Ancient wisdom emphasizes the importance of protecting the environment and living sustainably, which is critical in the face of global environmental crises.

- (c) **Value of holistic thinking:** Ancient knowledge systems, like those in India, integrate various fields of study, such as science, philosophy, and art. This approach can inspire interdisciplinary solutions to modern problems.
- (d) **Learning from the past:** By studying ancient knowledge, we can avoid repeating mistakes made in history and use successful practices to address modern issues.

Ancient knowledge is not just about history—it is a treasure of wisdom that remains relevant in shaping modern civilization. It has given us the foundation for science, art, governance, and philosophy. By understanding and applying this knowledge, we can build a more balanced, sustainable, and ethical society while staying connected to our roots. Ancient wisdom serves as a guide to harmonize tradition with progress for a better future.

1.4 INDIAN PHILOSOPHICAL SYSTEMS

Development and distinctive features of vedic and non-vedic philosophical systems:

Indian philosophy is one of the oldest and most diverse traditions of thought in the world. It developed over thousands of years, reflecting the cultural, spiritual, and intellectual growth of India. Indian philosophy can be broadly divided into two categories: **Vedic (orthodox)** and **non-Vedic (heterodox)** systems. These systems explore questions about life, the universe, knowledge, and liberation (moksha), while offering diverse perspectives on human existence. Below is a detailed explanation of their development and features in simple terms.

1.4.1 Development of Indian Philosophical Systems

(a) Vedic Philosophical Systems

The Vedic systems are based on the teachings of the Vedas, which are the oldest sacred texts in Indian culture. These systems accept the authority of the Vedas and are also known as *astika* (orthodox) schools. Over time, they developed into six major philosophical schools (*shad-darshanas*):

1. **Nyaya:** Focused on logic and reasoning to gain knowledge and understand reality.
2. **Vaisheshika:** Emphasized the study of matter and atoms to understand the universe.
3. **Samkhya:** Explained the duality of consciousness (*Purusha*) and matter (*Prakriti*).
4. **Yoga:** Focused on physical, mental, and spiritual practices to achieve self-realization.
5. **Mimamsa:** Stressed the importance of rituals and ethics as described in the Vedas.
6. **Vedanta:** Focused on understanding the ultimate reality (*Brahman*) and liberation (*moksha*).

These systems emerged between 1500 BCE and 500 CE, reflecting the evolving spiritual and intellectual needs of Indian society.

(b) Non-vedic philosophical systems:

Non-Vedic systems do not accept the authority of the Vedas. They are also called *nastika* (heterodox) schools and include:

1. **Buddhism:** Founded by Gautama Buddha, it emphasizes the Four Noble Truths and the Eightfold Path to achieve enlightenment (*nirvana*).
2. **Jainism:** Founded by Mahavira, it promotes non-violence (*ahimsa*), self-discipline, and the path to liberation through ethical living.
3. **Charvaka:** A materialistic philosophy that rejects the idea of the soul, afterlife, and religious rituals.

These systems arose between 600 BCE and 200 BCE, often as a response to the rigid ritualism and social hierarchies of Vedic traditions.

1.4.2 Distinctive Features of Vedic Philosophical Systems

1. **Authority of the vedas:** Vedic systems consider the Vedas as the ultimate source of knowledge and guidance. They base their teachings on Vedic hymns, rituals, and philosophies.

2. **Focus on liberation (moksha):** The goal of life is to attain liberation from the cycle of birth and death (*samsara*) by realizing the ultimate reality (*Brahman* or self).
3. **Belief in atman (self):** Most Vedic systems, especially Vedanta, emphasize the concept of the *atman* (individual soul) as eternal and connected to the universal reality (*Brahman*).
4. **Integration of rituals and philosophy:** Vedic systems combine spiritual rituals with philosophical inquiry, as seen in Mimamsa, which stresses the importance of performing Vedic rituals.
5. **Diversity of thought:** Although they all accept the Vedas, the six schools differ in their approach. For example, Nyaya focuses on logic, while Yoga emphasizes discipline and meditation.

1.4.3 Distinctive Features of Non-Vedic Philosophical Systems

1. **Rejection of vedic authority:** Non-vedic systems do not accept the Vedas as a source of knowledge. Instead, they rely on personal experience, reasoning, and practical teachings.
2. **Focus on practical ethics:** Buddhism and Jainism emphasize ethical behaviour, compassion, and non-violence over rituals. They provide a practical path to improve one's life and achieve liberation.
3. **Diverse Perspectives on Liberation:**
 - **Buddhism:** Liberation is achieved by overcoming desires and ignorance through meditation and ethical living.
 - **Jainism:** Liberation comes through self-discipline, non-violence, and detachment from material possessions.
 - **Charvaka:** Rejects the concept of liberation, focusing instead on enjoying life in the present.
4. **Human-centered philosophy:** Non-vedic systems place more emphasis on human effort and reason than on divine intervention or rituals.

5. **Social reform:** Many non-Vedic systems, like Buddhism and Jainism, challenged the caste system and ritualistic practices, promoting equality and simplicity.

1.4.4 Historical and Cultural Contexts

1. Vedic systems:

- The Vedic systems developed in a society deeply rooted in rituals, social hierarchies, and reverence for sacred texts.
- As Indian society evolved, these systems adapted to include more philosophical and spiritual elements, moving beyond rituals to focus on deeper questions about life and the universe.

2. Non-vedic systems:

- Non-vedic systems emerged during a time of social and cultural change, around 600 BCE. This was a period marked by dissatisfaction with rigid rituals, inequality, and the growing complexity of Vedic practices.
- These systems offered alternative paths to knowledge and liberation, focusing on simplicity, ethics, and direct experience.

1.4.5 Key Differences Between Vedic and Non-Vedic Systems

Aspect	Vedic Systems	Non-Vedic Systems
Authority	Accept the Vedas	Reject the Vedas
Focus	Rituals, philosophy, liberation	Practical ethics, equality, liberation
View on Atman	Belief in eternal soul (<i>atman</i>)	Mixed views (e.g., Buddhism denies the soul)
Social Structure	Support traditional hierarchies	Promote equality and reform
Path to Liberation	Knowledge, rituals, meditation	Ethics, self-discipline, reasoning.

Indian philosophical systems, both Vedic and non-Vedic, offer diverse and profound insights into life, ethics, and the universe. While Vedic systems provide a foundation rooted in tradition and spiritual inquiry, non-Vedic systems challenge rigid practices and focus on practical ways to improve life. Together, these systems reflect the richness of Indian thought and continue to inspire people around the world.

1.5 ANCIENT EDUCATIONAL INSTITUTIONS: TAKSASILA AND NALANDA

India has a long tradition of education and knowledge-sharing. Two of the most famous ancient educational institutions are **Takṣaśilā** and **Nālandā**, which were not only centres of learning but also played a key role in spreading knowledge across Asia and the world. Here is a detailed note in simple English about these institutions and their notable alumni.

1.5.1 Taksasila

- Takṣaśilā was one of the earliest universities in the world, located in present-day Pakistan, near Rawalpindi.
- It existed around 1000 BCE and flourished until the 5th century CE.
- It was an important center for learning, attracting students from India, Persia, Greece, and China.

Subjects Taught

- Takṣaśilā offered a wide range of subjects, including:
 - Vedas and scriptures
 - Grammar and language
 - Philosophy and ethics
 - Medicine (Ayurveda)
 - Mathematics
 - Astronomy
 - Political science and law
 - Military science

Features:

- Education was informal, with no rigid curriculum. Students studied under renowned teachers (*gurus*).
- There were no fixed classrooms. Learning happened through discussions and debates.
- Students stayed until they mastered their chosen subject, which could take several years.

Notable alumni:**1. Panini:**

- A famous Sanskrit grammarian, Panini wrote the *Ashtadhyayi*, the earliest and most comprehensive grammar of Sanskrit. His work laid the foundation for linguistic studies.

2. Kautilya (Chanakya):

- The teacher, economist, and political strategist who authored *Arthashastra*, a treatise on statecraft, governance, and economics.
- He was the mentor of Chandragupta Maurya, the founder of the Maurya Empire.

3. Jivaka:

- A renowned physician who practiced Ayurveda. He became the personal doctor of Lord Buddha and the Magadha royal family.

1.5.2 Nālandā

- Nālandā was one of the world's first residential universities, located in present-day Bihar, India.
- It was established in the 5th century CE during the Gupta Empire and flourished until the 12th century CE.
- Nālandā attracted scholars from India, Tibet, China, Korea, and Southeast Asia.

Subjects taught:

- Nalanda offered a wide variety of subjects, such as:
 - Buddhist philosophy and theology
 - Logic and metaphysics
 - Grammar and language
 - Medicine
 - Mathematics
 - Astronomy
 - Arts and architecture

Features:

- The university had an organized structure with hostels, libraries, and lecture halls.
- It housed over 10,000 students and 2,000 teachers.
- Nālandā's library, known as *Dharmaganja*, was one of the largest in the world, containing thousands of manuscripts.
- It had a formal admission process, and students had to prove their merit to enrol.

Notable alumni:**1. Aryabhata:**

- The famous mathematician and astronomer who studied at Nālandā. He made significant contributions, including the decimal system, and theories about planetary motion.

2. Xuanzang (Hiuen Tsang):

- A Chinese monk and scholar who studied at Nālandā during the 7th century. He later translated Buddhist texts into Chinese and wrote about Indian culture and education.

3. Dharmapala:

- A Buddhist scholar who later became the head of Nālandā. He contributed to Buddhist philosophy and spread its teachings across Asia.

4. Shilabhadra:

- A celebrated Buddhist philosopher and teacher who mentored Xuanzang and played a key role in spreading Mahayana Buddhism.

1.5.3 Contributions to the Spread of Knowledge

Takṣaśilā

- Takṣaśilā was a melting pot of cultures and knowledge systems. It spread Indian knowledge in areas like medicine, linguistics, and political science to neighboring regions like Persia and Greece.
- The graduates of Takṣaśilā, like Kautilya and Panini, created works that influenced global intellectual traditions.

Nālandā:

- Nālandā played a key role in spreading Buddhism across Asia. Scholars like Xuanzang and Dharmapala carried Buddhist teachings to countries like China, Japan, and Korea.
- Its advanced system of education and vast library set a model for future universities around the world.

1.5.4 Historical and Cultural Significance

- Both Takṣaśilā and Nālandā showcased India's deep commitment to education and intellectual growth.
- These institutions emphasized interdisciplinary learning, combining science, philosophy, and spirituality.
- They fostered cultural exchange, as students and teachers from different countries brought new ideas and innovations back to their homelands.

1.5.5 Decline

- Takṣaśilā declined after the 5th century CE due to invasions and political instability in the region.
- Nālandā was destroyed in the 12th century CE by Bakhtiyar Khilji, a Turkish invader, who burned its library and ended its legacy as a global centre of learning.

1.5.6 Legacy

Despite their decline, the contributions of Takṣaśilā and Nālandā continue to inspire modern education. Their emphasis on holistic learning, cultural exchange, and intellectual freedom serves as a model for universities today. These ancient institutions remind us of India's rich heritage as a centre of global knowledge.

Exercise

(A) Multiple Choice Questions (MCQs):

- What is the primary goal of Indian Knowledge Systems (IKS)?
 - Accumulating facts
 - Achieving self-realization and liberation**
 - Gaining material wealth
 - Mastering Sciences and Arts
- Which of the following is NOT a subject taught at Takṣaśilā?
 - Vedas and Scriptures
 - Medicine (Ayurveda)
 - Computer Science**
 - Military Science
- Which Indian philosopher is known for writing the "Ashtadhyāyī"?
 - Kautilya
 - Panini**
 - Aryabhata
 - Jivaka
- Which Indian educational institution was destroyed in the 12th century CE by Bakhtiyar Khilji?
 - Takṣaśilā
 - Nālandā**
 - University of Delhi
 - Aligarh Muslim University
- Which of the following is a feature of the Vedic philosophical systems?
 - Rejection of rituals
 - Emphasis on ethics alone
 - Acceptance of the authority of the Vedas**
 - Focus on human-centered philosophy

6. Which of the following is NOT an achievement of Aryabhata?
- (a) Concept of place value system
 - (b) Theories on planetary motion
 - (c) Development of the decimal system
 - (d) Founder of Jainism**
7. What is the primary difference between Vedic and Non-vedic systems of philosophy?
- (a) Vedic systems focus on personal experience, while Non-vedic systems emphasize rituals.
 - (b) Non-vedic systems reject the authority of the Vedas.**
 - (c) Vedic systems are materialistic in nature.
 - (d) Non-vedic systems believe in a cyclical concept of time.
8. Which Indian institution had the Dharmaganja library, one of the largest in the world at its time?
- (a) Takṣaśilā
 - (b) Nālandā**
 - (c) University of Nalanda
 - (d) Patanjali Academy
9. Which philosophy emphasized non-violence (ahimsa) and self-discipline for liberation?
- (a) Jainism**
 - (b) Buddhism
 - (c) Yoga
 - (d) Nyaya
10. What is the concept of 'moksha' in Indian philosophy?
- (a) Liberation from the material world**
 - (b) Accumulation of wealth
 - (c) Reincarnation
 - (d) Enlightenment

(B) Short Answer Questions:**(1 Mark)**

1. What is the main goal of the Indian Knowledge Systems (IKS)?
2. Which Indian philosopher wrote the *Arthashastra*?
3. What is the significance of the concept of 'moksha' in Indian philosophy?
4. Which ancient Indian institution attracted students from China, Korea, and Tibet?
5. Who was the ancient Indian scholar that authored the *Ashtadhyayi* on Sanskrit grammar?

(C) Short Answer Questions:**(2 Marks)**

1. Explain the significance of ancient knowledge in shaping modern civilization.
2. What were the main subjects taught at Takṣaśilā?
3. How did the Bodhayana Sutra contribute to the development of mathematics in ancient India?
4. Describe the importance of Ayurveda in Indian medical knowledge systems.
5. How did the concept of zero revolutionize mathematics and scientific thought?

(D) Long Answer Questions:**(3 Marks)**

1. Describe the distinctive features of Vedic and Non-vedic philosophical systems.
2. How did the institution of Nālandā contribute to the spread of knowledge?
3. What were Aryabhata's significant contributions to mathematics and astronomy?
4. Discuss the role of ancient Indian practices like sustainable agriculture and water conservation.
5. Explain the role of guru-shishya tradition in the Indian Knowledge System.

(2) Learning Objectives (Qualitative)

(3 Marks)

1. Explain the development and evolution of knowledge and how it is influenced by culture.
2. Discuss the role of technology and innovation in the growth of knowledge and education in modern times.
3. Discuss the impact of knowledge on society.
4. Explain the importance of knowledge and its application in life.
5. Explain the significance of knowledge in education.
